

Parameter	Default cut-offs	Remedial action
Per Base Sequence Quality: assesses the quality scores across all bases at each position in the reads.	Warning: lower quartile for any base is less than 10, or if the median for any base is less than 25. Fail: lower quartile for any base is less than 5 or if the median for any base is less than 20.	Quality and adapter trimming. Transient quality dips or variable read lengths can also trigger errors, requiring careful evaluation rather than trimming.
Per Sequence Quality Scores: assesses the average quality score per read.	Warning: mean quality is below 27, equates to a 0.2% error rate. Fail: mean quality is below 20, equates to a 1% error rate.	Quality trimming may help. Complex distributions should be checked against per-tile quality for underlying issues.
Per Base Sequence Content: measures the proportion of each base (A, T, G, C) at each position.	Warning: difference between A and T, or G and C is greater than 10% in any position. Fail: difference between A and T, or G and C is greater than 20% in any position.	Some deviation is tolerable. Severe imbalance early in reads may indicate adapter/primer artifacts.
Per Sequence GC Content: plots the distribution of GC content across all reads.	Warning: if the sum of the deviations from the normal distribution represents $\geq 15\%$ of the reads. Fail: if the sum of the deviations from the normal distribution represents $\geq 30\%$ of the reads.	Major deviations may indicate contamination or bias. Sharp peaks suggest specific contaminants (adapter dimers), while broad peaks may indicate cross-species contamination.
Per Base N Content: shows the percentage of ambiguous base calls (N) at each base position.	Warning: if any position shows an N content of $>5\%$. Fail: if any position shows an N content of $>20\%$.	5% N at any base position may require investigation or trimming. High proportions of Ns usually indicate quality loss and should be assessed alongside quality metrics.
Sequence Length Distribution: displays the distribution of read lengths.	Warning: all sequences are not the same length. Fail: any of the sequences have zero length.	For some sequencing platforms it is entirely normal to have different read lengths so warnings here can be ignored.
Sequence Duplication Levels: determines the extent of duplication in the dataset.	Warning: non-unique sequences make up $>30\%$ of the total. Fail: non-unique sequences make up $>50\%$ of the total.	Duplicates can be removed in downstream analyses. They may arise from PCR artifacts (technical) or random sampling (biological); both appear identical at the sequence level and are reported together.
Overrepresented Sequences: flags sequences that occur more often than expected.	Warning: any sequence found to represent $>0.1\%$ of the total. Fail: any sequence found to represent $>1\%$ of the total.	Investigate for adapters, primers, contamination, or biological interest. Trim as necessary.
Adapter Content: identifies if known adapter sequences are present.	Warning: any sequence is present in $>5\%$ of all reads. Fail: any sequence is present in $>10\%$ of all reads.	Adapter trimming.
K-mer Content: highlights overrepresented short sequences.	Warning: any k-mer imbalanced with a binomial p-value <0.01 . Fail: any k-mer imbalanced with a binomial p-value $<10^{-5}$	Investigate if specific k-mers correspond to adapters, rRNA or primers and trim if necessary.